

**DEPARTMENT OF COMPUTER ENGINEERING,
FACULTY OF ENGINEERING, UNIVERSITY OF JAFFNA
EC4010 – DIGITAL IMAGE PROCESSING
LABORATORY SESSION 6**

Analysis and design of asynchronous sequential circuits (Part 2)

(Time duration 3 hours only)

1. OBJECTIVES :

- Structural modeling asynchronous sequential circuits
- Determining outputs of an asynchronous sequential circuit for a given sequence of inputs
- Identify stability considerations
- Identify and mitigate race conditions

2. SOFTWARE:

- Quartus Prime Lite Edition
- ModelSim - Intel FPGA Starter Edition Model Technology ModelSim

3. LABORATORY TASKS :

Q1. Modeling of an asynchronous sequential circuit.

- a) Derive the transition table of the asynchronous circuit given below.

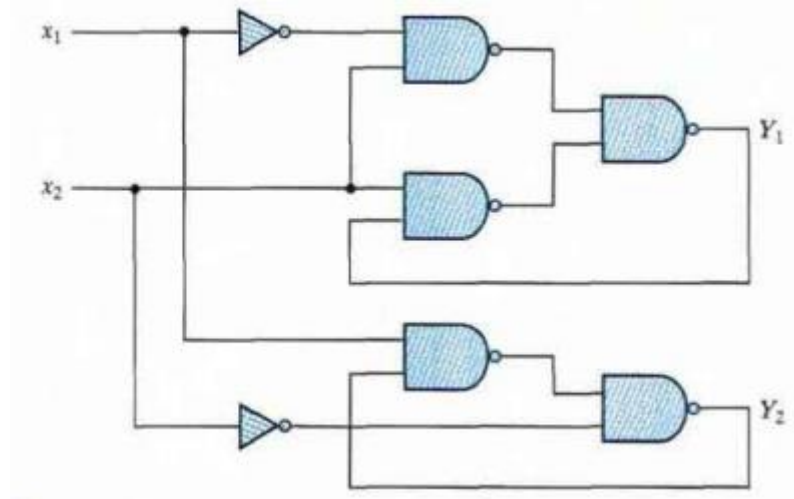


Figure 1: Asynchronous Circuit 1

- b) Model the circuit using Verilog and obtain a diagram of it using the RTL viewer. Assume all gates and feedback paths consume 10 ps.
- c) Create a testbench and determine the sequence of internal states for the sequence of inputs. x_1x_2 : 00,10,11,01,11,10,00.
- d) Does the output obtained in c) correspond with the transition table obtained in part a), explain.

Q2. Use the diagram given below to answer the following questions.

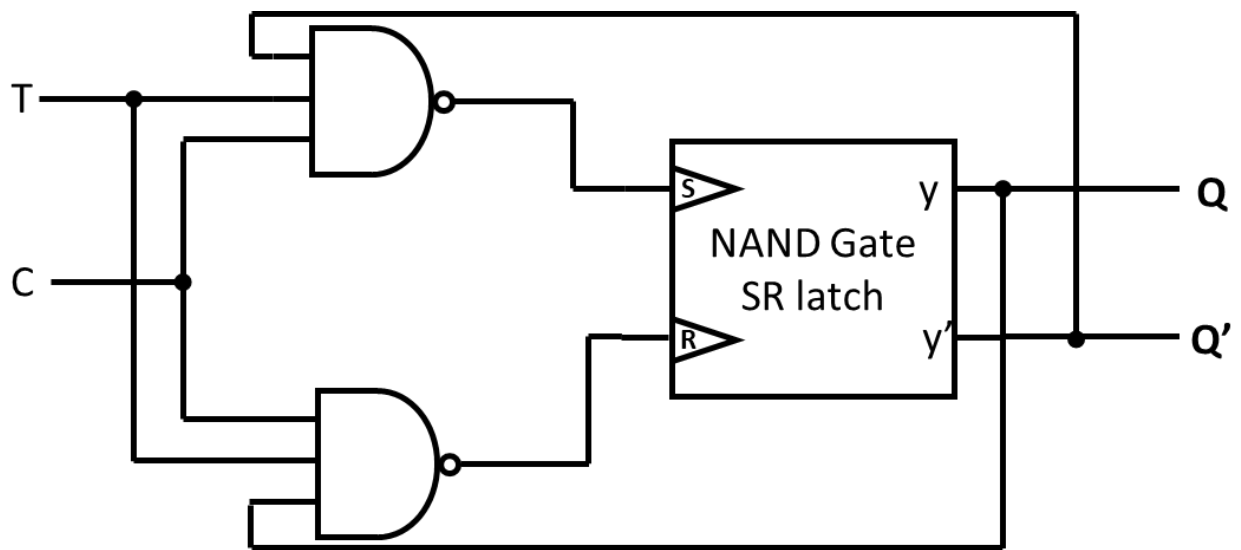


Figure 2: Asynchronous circuit 2

- Derive the equations for S and R; use them to draw the maps for S and R.
- Use the truth-table of a NAND gate SR Latch and part a) to draw the transition table for the above circuit.
- Using the given Verilog files for the circuit and testbench observe the Gate level simulation (simulate for 320 ns).
- Explain any stability considerations observed (if any) using the transition table obtained in b).

Q3. Given below is are circuit diagrams of a negative edge triggered T flip-flop.

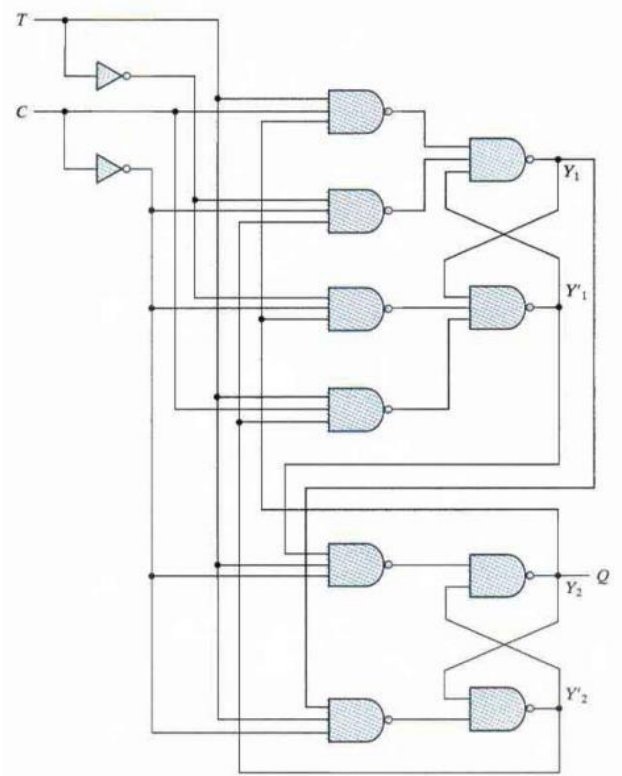


Figure 3: Negative edge triggered T flip-flop

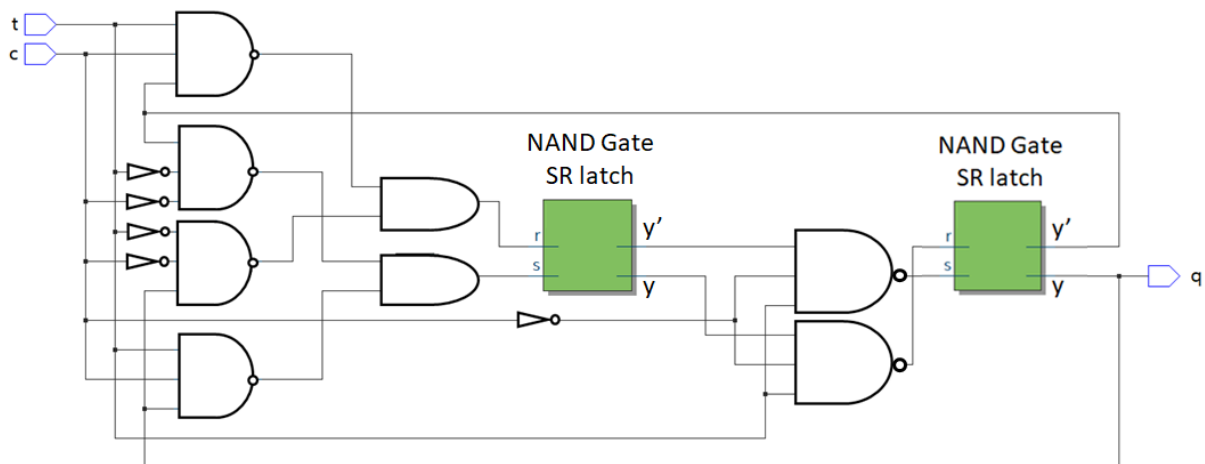


Figure 4: Negative edge triggered T flip-flop with SR latches

- Construct the circuit with SR latches in Verilog. Assume all gates and feedback paths consume 10 ps and the SR latch consumes no time.
- Using the same testbench provided for Q2 observe the gate level simulation (simulate for 320 ns).
- Explain how the gate level simulation follows the functionality of a negative edge triggered T flip-flops.

4. Submission instructions:

- **Submit a report**
- **Submit Verilog files for models and testbench, also submit testvectors if any.**
- **Use the given folder structure**